

Computer Network Defense: Assignment

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1 Instructions

Discuss each lab's questions with your group at the end of that lab, with respect to the tools used within the lab. After the final lab, submit a single PDF per group on Canvas that answers every question below, organized by lab, in a succinct, but complete, manner.

2 Firewall Lab

Discuss the following with your group and provide a brief response to each.

1. What is the difference between **denying** traffic from a specific IP and **deleting** a rule that allows a service?
2. In a real organization, who would be responsible for managing firewall rules? What process might they follow?
3. The Linux port-based approach and the Windows application-based approach each have trade-offs. Which do you think is easier to maintain? Which do you think is more secure?

3 Docker Lab

Discuss the following with your group and provide a brief response to each.

1. You SSH'd into the container as a non-root user. What risks would need to be considered if you had SSH'd into the container as root?
2. A "container escape" is when a process inside a container gains access to the host. Based on what you observed in this lab, what features or configurations do you think would make a container escape more likely?
3. When you removed the container, everything inside it was gone except for files in the shared volume. How could this property be useful for security testing or incident response?

4 Wireshark Lab

Discuss the following with your group and provide a brief response to each.

1. Wireshark can only capture traffic that reaches your network interface. On a modern switched network, you typically only see your own traffic. What technique would an attacker need to use to capture traffic meant for someone else on the same network?
2. You set up a firewall in a previous lab. Does a firewall prevent someone on the same network from sniffing your traffic? Why or why not?
3. You used Wireshark in a defensive context here by observing your own machine. What would it look like to use this tool offensively, and what authorization would you need to do so legally?

5 Detection Lab

Discuss the following with your group and provide a brief response to each.

1. fail2ban both detects an attack and responds to it automatically. What are the advantages and risks of responding automatically instead of only alerting a person?
2. An attacker guesses one password every hour from each of 1,000 different IP addresses. Would fail2ban stop this? What kind of system or approach would be needed to notice it?
3. Your firewall, isolation, and monitoring are all layers of **defense in depth**. Describe a scenario where an attacker gets past one layer but is stopped or detected by another.